

4.2 Null Spaces, Column Spaces, Row Spaces, and Linear Transformations

I

The Null Space of a Matrix

Definition

The null space of an $m \times n$ matrix A , written as $\text{Nul } A$, is the set of all solutions of the homogenous equation $A\vec{x} = \vec{0}$. In set notation,

$$\text{Nul } A = \{ \vec{x} : \vec{x} \text{ is in } \mathbb{R}^n \text{ and } A\vec{x} = \vec{0} \}$$

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Example 1

Let $A = \begin{bmatrix} 1 & -3 & -2 \\ 5 & 9 & 1 \end{bmatrix}$ and $\vec{u} = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$. Determine if $\vec{u}$ belongs to the null space of A .

Solution

Compute $A\vec{u}$ to see if it equals $\vec{0}$.

$$\begin{bmatrix} 1 & -3 & -2 \\ 5 & 9 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

thus, $\vec{u}$ belongs to $\text{Nul } A$.

Theorem 2

The null space of $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions to a system $A\vec{x} = \vec{0}$ of m homogenous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

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Proof

Certainly $\text{Nul } A$ is a subspace of $\mathbb{R}^n$ because A has n columns. We must show that $\text{Nul } A$ satisfies the three properties of a subspace as defined in 4.1.

Let $\vec{u}$ and $\vec{v}$ represent any two vectors in $\text{Nul } A$. Then

$$A\vec{u} = \vec{0} \text{ and } A\vec{v} = \vec{0}$$

Now, using properties of matrix operations

$$A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v} = \vec{0} + \vec{0} = \vec{0}$$

and for scalars

$$A(c\vec{u}) = c(A\vec{u}) = c(\vec{0}) = \vec{0}$$

which shows that $\text{Nul } A$ is a subspace of $\mathbb{R}^n$.

Example 2

Let H be the set of all vectors in $\mathbb{R}^4$ whose coordinates a, b, c, d satisfy the equations $a - 2b + 5c = d$ and $c - a = b$. Show that H is a subspace of $\mathbb{R}^4$.

Solution

Rearrange the equations that describe the elements of H , and note that H is the set of all solutions of the following system of homogeneous linear equations:

$$\begin{aligned} a - 2b + 5c - d &= 0 \\ -a - b + c &= 0 \end{aligned}$$

By theorem 2, H is a subspace of $\mathbb{R}^4$.

An explicit description of $\text{Nul } A$

There is no obvious relation between vectors in $\text{Nul } A$ and the entries in A . (I love my wife ❤️) We say that $\text{Nul } A$ is defined implicitly, because it is defined by a condition that must be checked. No explicit list or description of the elements in $\text{Nul } A$ is given. However, solving the equation $A\vec{x} = \vec{0}$ amounts to producing an explicit description of $\text{Nul } A$.

Example 3

Find a spanning set for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Solution

The first step is to find the general solution of $A\vec{x} = \vec{0}$ in terms of free variables. We do this by performing row operations on $[A \ \vec{0}]$ to reduced echelon form.

$$A \sim \left[\begin{array}{ccccc|c} \textcircled{1} & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & \textcircled{1} & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} x_1, x_3 \text{ - pivots} \\ x_2, x_4, x_5 \text{ - free} \end{array}$$

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5$

Writing out the system in parametric form:

$$\begin{aligned} x_1 - 2x_2 + 0x_3 - 1x_4 + 3x_5 &= 0 \\ 0x_1 + 0x_2 + 1x_3 + 2x_4 - 2x_5 &= 0 \end{aligned}$$

This gives:

$$\begin{aligned} x_1 &= 2x_2 + x_4 - 3x_5 & x_2 &= r, \quad x_4 = s, \quad x_5 = t \\ x_3 &= -2x_4 + 2x_5 \end{aligned}$$

$$\vec{x} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = r \underbrace{\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_{\vec{u}} + s \underbrace{\begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}}_{\vec{v}} + t \underbrace{\begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}}_{\vec{w}} \quad \%$$

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III

Every linear combination of $\vec{u}$, $\vec{v}$ and $\vec{w}$ is an element of $\text{Nul } A$ and vice versa. Thus $\{\vec{u}, \vec{v}, \vec{w}\}$ is a spanning set for $\text{Nul } A$.

Two points should be made about the solution of Example 3 that apply to all problems of this type where $\text{Nul } A$ contains nonzero vectors. These facts will be used later.

1. The spanning set produced by the method in Example 3 is automatically linearly independent because the free variables are the weights on the spanning vectors. For instance look at the 2nd, 4th and 5th entries in the solution vector in

$$x_2 \vec{u} + x_4 \vec{v} + x_5 \vec{w}$$

and note that it can be zero only if the weights are all zero.

2. When $\text{Nul } A$ contains nonzero vectors, the number of vectors in the spanning set for $\text{Nul } A$ equals the number of free variables in the equation $A\vec{x} = \vec{0}$.

The Column Space of a Matrix

Another important subspace associated with a matrix is its column space. Unlike the null space, the column space is defined explicitly via linear combinations.

Definition

The column space of an $m \times n$ matrix A , written as $\text{Col } A$, is the set of all linear combinations of A . If $A = [a_1, \dots, a_n]$, then

$$\text{Col } A = \text{Span}\{a_1, \dots, a_n\}$$

Since $\text{Span}\{a_1, \dots, a_n\}$ is a subspace, by theorem 1, the next theorem follows from the definition of $\text{Col } A$ and the fact that the columns of A are in $\mathbb{R}^m$.

Theorem 3

The column space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^m$.

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Note that a typical vector in $\text{Col } A$ can be written as $A\vec{x}$ for some $\vec{x}$ because the notation $A\vec{x}$ stands for a linear combination of the columns of A . That is,

$$\text{Col } A = \{\vec{b} : \vec{b} = A\vec{x} \text{ for some } \vec{x} \text{ in } \mathbb{R}^n\}$$

The notation $A\vec{x}$ for vectors in $\text{Col } A$ also shows that $\text{Col } A$ is the range of the linear transformation $\vec{x} \mapsto A\vec{x}$. We will return to this point of view at the end of the section.

Example 4

Find a matrix A such that $W = \text{Col } A$.
(W is a set of vectors)

$$W = \left\{ \begin{bmatrix} 6a-b \\ a+b \\ -7a \end{bmatrix} : a, b \text{ in } \mathbb{R} \right\}$$

Solution

First, write W as a set of linear combinations.

$$W = \left\{ a \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} : a, b \text{ in } \mathbb{R} \right\} = \text{Span} \left\{ \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Second, use the vector set as columns of A , $A = \begin{bmatrix} 6 & -1 \\ 1 & 1 \\ -7 & 0 \end{bmatrix}$

Then $W = \text{Col } A$, as desired.

Recall from theorem 4 in section 1.4 that the columns of A span $\mathbb{R}^m$ if and only if the equation $A\vec{x} = \vec{b}$ has a solution for each $\vec{b}$. We can restate this fact as follows:

The column space of an $m \times n$ matrix A is all of $\mathbb{R}^m$ if and only if the equation $A\vec{x} = \vec{b}$ has a solution for each $\vec{b}$ in $\mathbb{R}^m$.

The Row Space

If A is an $m \times n$ matrix, each row has n entries and thus can be identified with a vector in $\mathbb{R}^n$. The set of all linear combinations of the row vectors is called the row space of A and is denoted by $\text{Row } A$. Each row has n entries, so $\text{Row } A$ is a subspace of $\mathbb{R}^n$. Since the rows of A are identified with the columns of A^T , we could also write $\text{Col } A^T$ in place of $\text{Row } A$.

Example 5

$$\text{Let } A = \begin{bmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{bmatrix} \quad \begin{aligned} r_1 &= (-2, -5, 8, 0, -17) \\ r_2 &= (1, 3, -5, 1, 5) \\ r_3 &= (3, 11, -19, 7, 1) \\ r_4 &= (1, 7, -13, 5, -3) \end{aligned}$$

$$\text{Row } A = \text{Span}\{r_1, r_2, r_3, r_4\}$$

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The Contrast Between Nul A and Col A

It is natural to wonder how the null space and the column space of a matrix are related. In fact, the two spaces are quite dissimilar, as examples 6-8 will show. Nevertheless, a surprising connection between the null space and the column space will emerge in section 4.5 after more theory is available.

Example 6

Let $A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$

- If the column space of A is a subspace of $\mathbb{R}^k$, what is k ?
- If the null space of A is a subspace of $\mathbb{R}^k$, what is k ?

Solution

- 3, the columns of A have 3 entries
- 4, $\text{Nul } A$ is a subspace of $\mathbb{R}^n$ ($m \times n$)

Example 7

With A as in Example 6, find a nonzero vector in $\text{Col } A$, and a nonzero vector in $\text{Nul } A$.

Solution

A vector in $\text{Col } A$ is easy to find, any of the four column vectors will do

$$\vec{c}_1 = \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}, \vec{c}_2 = \begin{bmatrix} 4 \\ -5 \\ 7 \end{bmatrix}, \vec{c}_3 = \begin{bmatrix} -2 \\ 7 \\ -8 \end{bmatrix}, \vec{c}_4 = \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix}$$

To find a vector in $\text{Nul } A$, we need to solve the equation $A\vec{x} = \vec{0}$, the null space of A is the set of all solutions to this homogeneous equation. setting up the augmented matrix and performing row operations.

$$\begin{array}{l} \begin{bmatrix} 2 & 4 & -2 & 1 & | & 0 \\ -2 & -5 & 7 & 3 & | & 0 \\ 3 & 7 & -8 & 6 & | & 0 \end{bmatrix} \xrightarrow{\begin{array}{l} r_3 \leftarrow r_3 - 2r_1 \\ r_2 \leftarrow r_2 + r_1, r_3 \leftarrow r_3 - r_2 \end{array}} \begin{bmatrix} 2 & 4 & -2 & 1 & | & 0 \\ 0 & -1 & 5 & 4 & | & 0 \\ 1 & 3 & -6 & 5 & | & 0 \end{bmatrix} \xrightarrow{\begin{array}{l} r_3 \leftarrow r_3 - 2r_2 \\ r_1 \leftrightarrow r_3 \end{array}} \begin{bmatrix} 1 & 3 & -6 & 5 & | & 0 \\ 0 & -1 & 5 & 4 & | & 0 \\ 0 & 0 & 0 & -17 & | & 0 \end{bmatrix} \xrightarrow{r_1 \leftarrow r_1 + 3r_2} \begin{bmatrix} 1 & 0 & 9 & 17 & | & 0 \\ 0 & -1 & 5 & 4 & | & 0 \\ 0 & 0 & 0 & -17 & | & 0 \end{bmatrix} \end{array}$$

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$$\begin{aligned} r_1 &\leftarrow r_1 + r_3, \quad r_2 \leftarrow r_2 \cdot (-1) \\ r_3 &\leftarrow r_3 \cdot (-1/17) \end{aligned}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 9 & 0 & 0 \\ 0 & 1 & -5 & -4 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array} \right]$$

x_1, x_2, x_4 = pivot columns

x_3 = free variable column

$$x_1 + 0x_2 + 9x_3 + 0x_4 = 0$$

$$x_2 - 5x_3 + 0x_4 = 0$$

$$x_4 = 0$$

$$\left[\begin{array}{cccc|c} \textcircled{1} & 0 & 9 & 0 & 0 \\ 0 & \textcircled{1} & -5 & 0 & 0 \\ 0 & 0 & 0 & \textcircled{1} & 0 \end{array} \right]$$

$x_1 \ x_2 \ x_3 \ x_4$

$$x_2 = 5x_3$$

$$x_1 = -9x_3$$

$$\vec{x} = \begin{bmatrix} -9x_3 \\ 5x_3 \\ x_3 \\ 0 \end{bmatrix}, \quad x_3 = s \Rightarrow \vec{x} = s \begin{bmatrix} -9 \\ 5 \\ 1 \\ 0 \end{bmatrix} = s \vec{u}$$

$\vec{u}$ is a nonzero vector in $\text{Nul } A$.

Example 8

With A as in example 6, let $\vec{u} = \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}$

a. Determine if $\vec{u}$ is in $\text{Nul } A$. Could $\vec{u}$ be in $\text{Col } A$?

b. Determine if $\vec{v}$ is in $\text{Col } A$. Could $\vec{v}$ be in $\text{Nul } A$?

Solution

a. $\text{Nul } A$ is a subspace of $\mathbb{R}^4$, $\text{Col } A$ of $\mathbb{R}^3$. Since $\vec{u}$ is in $\mathbb{R}^4$, it can't be in $\text{Col } A$.

To see if $\vec{u}$ is in $\text{Nul } A$, we can simply compute $A\vec{u}$ to see if it equals $\vec{0}$.

$$A\vec{u} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 6-8+2 \\ -6+10-7 \\ 9-14+8 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

b. Reduce $[A \ \vec{v}]$ to echelon form

$$\left[\begin{array}{cccc|c} 2 & 4 & -2 & 1 & 3 \\ -2 & -5 & 7 & 3 & -1 \\ 3 & 7 & -8 & 6 & 3 \end{array} \right] \sim \left[\begin{array}{cccc|c} 2 & 4 & -2 & 1 & 3 \\ 0 & 1 & -5 & -4 & -2 \\ 0 & 0 & 0 & 17 & 1 \end{array} \right]$$

System is clearly consistent.

$\vec{v}$ is in $\text{Col } A$.

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VII

contrast between $\text{Nul } A$ and $\text{Col } A$ for an $m \times n$ matrix A

Nul A

1. $\text{Nul } A$ is a subspace of $\mathbb{R}^n$
2. $\text{Nul } A$ is implicitly defined; that is, you are given only a condition ($A\vec{x} = \vec{0}$) that vectors in $\text{Nul } A$ must satisfy.
3. It takes time to find vectors in $\text{Nul } A$. Row operations on $[A \ \vec{0}]$ are required.
4. There is no obvious relation between $\text{Nul } A$ and the entries in A .
5. A typical vector $\vec{v}$ in $\text{Nul } A$ has the property $A\vec{v} = \vec{0}$.
6. Given a specific vector $\vec{v}$, it is easy to tell if $\vec{v}$ is in $\text{Nul } A$. Just compute $A\vec{v}$.
7. $\text{Nul } A = \{\vec{0}\}$ if and only if the equation $A\vec{x} = \vec{0}$ has only the trivial solution.
8. $\text{Nul } A = \{\vec{0}\}$ if and only if the linear transformation $\vec{x} \rightarrow A\vec{x}$ is one-to-one.

Col A

1. $\text{Col } A$ is a subspace of $\mathbb{R}^m$
2. $\text{Col } A$ is explicitly defined; that is, you are told how to build vectors in $\text{Col } A$.
3. It is easy to find vectors in $\text{Col } A$. The columns of A are displayed; others are formed from them.
4. There is an obvious relation between $\text{Col } A$ and the entries in A , since each column in A is in $\text{Col } A$.
5. A typical vector $\vec{v}$ in $\text{Col } A$ has the property that $A\vec{x} = \vec{v}$ is consistent.
6. Given a specific vector $\vec{v}$, it may take time to tell if $\vec{v}$ is in $\text{Col } A$. Row operations on $[A \ \vec{v}]$ is required.
7. $\text{Col } A = \mathbb{R}^m$ if and only if the equation $A\vec{x} = \vec{b}$ has a solution for every $\vec{b}$ in $\mathbb{R}^m$.
8. $\text{Col } A = \mathbb{R}^m$ if and only if the linear transformation $\vec{x} \rightarrow A\vec{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^m$.

Kernel and Range of a Linear Transformation

Subspaces of vector spaces other than $\mathbb{R}^n$ are often described in terms of a linear transformation instead of a matrix. To make this precise, we generalize the definition given in Section 1.8. (p. 94)

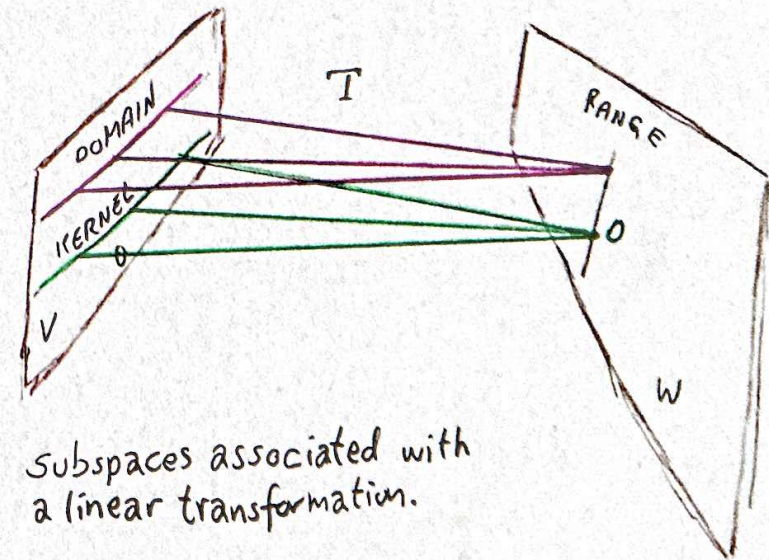
Definition

A linear transformation from a vector space V into a vector space W is a rule that assigns to each vector $\vec{x}$ in V a unique vector $T(\vec{x})$ in W , such that

- (i) $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v}$ in V , and
- (ii) $T(c\vec{u}) = cT(\vec{u})$ for all $\vec{u}$ in V and all scalars c .

The kernel (or null space) of such a T is the set of all $\vec{u}$ in V such that $T(\vec{u}) = \vec{0}$ (the zero vector in W). The range of T is the set of all vectors in W of the form $T(\vec{x})$ for some $\vec{x}$ in V . If T happens to arise as a matrix transformation - say, $T(\vec{x}) = A\vec{x}$ for some matrix A - then the kernel and the range of T are just the null space and the column spaces of A , as defined earlier.

It is not difficult to show that the kernel of T is a subspace of V . The proof is essentially the same as the one for theorem 2 (p. 236). Also, the range of T is a subspace of W .



Subspaces associated with a linear transformation.